Security Assurance for Modern Big Data APIs

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ITADATA 2026

Context: Big Data APIs & Security Assurance

- **Big Data Ecosystem:** pipelines of distributed services behind REST APIs

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- **Big Data Ecosystem:** pipelines of distributed services behind REST APIs
- **Trust Boundary:** API Gateways centralize main security controls
- **Assurance Gap:** infrastructure and functional checks miss API security semantics
- **Consequence:** an API-level weakness can expose the pipelines and data behind the interface

State of the Art Limitations and Objectives

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- **Syntactic Blindness:**
  valid requests can still violate security

Our Objectives

- **Semantic Verification:**
  check security behavior, not only syntax

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- **Syntactic Blindness:**
  valid requests can still violate security
- **The Oracle Problem:**
  no criterion to judge a violation
- **Shallow & Flaky Tools:**
  tools trade depth for reach, results vary across runs

Our Objectives

- **Semantic Verification:**
  check security behavior, not only syntax
- **Specified Oracles:**
  an explicit rule for each guarantee
- **Agnostic & Deterministic:**
  portable across targets, reproducible across runs

Our Approach: Overview

Core Idea: Derive valid tests from OpenAPI contracts and evaluate security behavior using specified, deterministic oracles.

- **Application Agnosticism & Contract-Driven Testing**
- **Eight-Domain Taxonomy & Privilege-Aware Coverage**
- **Specified Oracles for Security Guarantees**
- **Deterministic Assessment Engine**

Our Approach: Agnosticism and Contract-Driven

- **Application agnosticism:** no target-specific code; the target is described by its OpenAPI spec and configuration
- **Configuration as source of variation:** all target-specific values live in one file
- **Contract-driven testing:** the spec defines the attack surface and guides the construction of valid requests

Our Approach: Taxonomy and Privilege-Aware Coverage

8-Domain Taxonomy

ID	Domain
D0	API Discovery
D1	Identity & Authentication
D2	Authorization
D3	Data Integrity
D4	Availability & Resilience
D5	Observability
D6	Configuration & Hardening
D7	Business Logic & Sensitive Flows

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Privilege-Aware Box Gradient

- **Black:** *D1: does a protected endpoint refuse requests without credentials?*
- **Grey:** *D2: can user A read the resources of user B?*
- **White:** *D6: does the gateway enforce the HSTS security header?*

Our Approach: Specified Oracles

Every test has its own specified oracle, built in three steps:

1. Identify the guarantee → **2.** Define the violation → **3.** Encode the rule

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Domain	Criterion from	Violation
D1 Authentication	spec: endpoint protected	2xx without credentials
D2 Authorization	two roles compared	user A reads user B's resource
D6 Configuration	gateway configuration	HSTS header missing

Our Approach: Deterministic Assessment Engine

- **Dependency-aware scheduling:** a test runs only after its prerequisites
- **Reproducible assessment:** fixed configuration, order and oracles $\rightarrow$ same configuration, same verdict
- **Actionable output:** distinct exit codes separate a security violation from an infrastructure error

Experimental Setup

Target and gateway

- **Forgejo** 14.0.3 – **Kong** 3.9 DB-less
- PostgreSQL 15
- 3 accounts: admin + 2 users

Interface properties

shared by Big Data services behind a gateway

- Native OpenAPI specification
- Distinct privilege levels and parameterized endpoints
- Gateway configuration available

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The testbed combines:

- **Expected success:** e.g. security headers added by the gateway
- **Deliberately broken:** e.g. rate limiting disabled
- **Native behavior:** other outcomes come from Forgejo and Kong as deployed

Experimental Results

Two independent runs, same outcomes across **18 active tests in 7 domains**

9 PASS **7 FAIL** **2 SKIP** **0 ERROR**

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Key result: Contract vs. Behavior (D1)

Spec:	endpoints declared **protected**
Observed:	2xx without credentials
Verdict:	BYPASSED (74 findings)

Public by design: the contract is wrong $\rightarrow$ an analyst decides

Conclusions & Future Work

Contract-driven security assurance for Big Data APIs, with specified oracles

- Verifies security behavior, not only syntax
- Same outcomes across independent runs $\rightarrow$ ready for CI/CD
- Found a real contract–behavior discrepancy on Forgejo

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Future Work

- Automatic generation of security oracles
- Complete the coverage of the taxonomy
- Validation on other targets

Thanks! Questions?

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